Supplementary Material

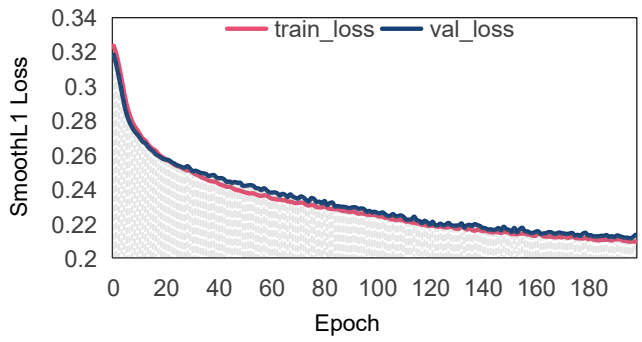


Fig. 1. **Training convergence.** Loss curves of the normal-dynamics evaluator.

I. Learned Normal-Dynamics Evaluator

This section details the normal-dynamics evaluator: its features, architecture, training, and capacity diagnostics. The network converts candidate trajectories into evidence of deviation from nominal dynamics rather than outputting recovery actions.

A. Input features and prediction task

The aligned flight logs are resampled to 50 Hz and converted into a 20-dimensional weakly phase-dependent feature vector. The features consist of first-order increments and local variation statistics, including position increments, velocity increments, speed-norm change, altitude change, acceleration increments, acceleration-norm change, angular-rate increments, angular-rate-norm change, displacement-increment norm, and short-window standard deviations of acceleration, angular rate, and speed increments. The supervised learning task is short-horizon feature prediction:

$$F_\theta : \mathbb{R}^{T\times d} \to \mathbb{R}^{K\times d}, \qquad d = 20,\ T = 50,\ K = 10.$$

Inputs and targets are robustly normalized using the median and interquartile range (IQR). Increment features emphasize local dynamic changes while reducing dependence on absolute position, mission phase, and trajectory geometry.

B. Network architecture

Table I specifies the normal-dynamics predictor. The model is a lightweight temporal convolutional predictor with two 1D convolutional layers, adaptive average pooling, and a two-layer multilayer perceptron (MLP) head. It contains 12,712 trainable parameters. This design prioritizes deterministic, low-cost inference: the online decision process requires a stable relative deviation measure, not a high-capacity trajectory generator.

Algorithm 1 Normal-dynamics learning and anomaly calibration

Require: Nominal flight logs D, history window T, prediction horizon K, feature dimension d
Ensure: Predictor F_θ, robust normalizer, nominal error statistics (m_e, q_e)
1: Resample logs to a 50 Hz time grid.
2: Extract the $[0, 14.0)$ s nominal segment and construct 20-dimensional increment/statistic features.
3: Fit a median/IQR robust normalizer on the nominal features.
4: Construct prediction samples (X_t, Y_t) with stride 5.
5: Train F_θ with Adam by minimizing $\mathrm{SmoothL1}(Y_t, F_\theta(X_t))$.
6: Compute nominal prediction errors e_t and estimate $m_e = \mathrm{median}(e_t)$ and $q_e = \mathrm{IQR}(e_t)$.
7: Freeze F_θ, the normalizer, and (m_e, q_e) for candidate-consequence evaluation.

TABLE I
Architecture of the Normal-Dynamics Predictor.

Module	Operation	Input	Output	Parameters
Input	Normalized feature window	50×20	50×20	0
Conv1	Conv1d(20,32,k=3,p=1)+ReLU	$20 \times T$	$32 \times T$	1,952
Conv2	Conv1d(32,32,k=3,p=1)+ReLU	$32 \times T$	$32 \times T$	3,104
Pooling	AdaptiveAvgPool1d(1)	$32 \times T$	32	0
MLP-1	Linear(32,32)+ReLU	32	32	1,056
MLP-2	Linear(32,200)	32	10×20	6,600
Total	Trainable parameters	–	–	**12,712**

C. Training data, loss function, and convergence

Training uses only the pre-fault nominal segments. This reconstruction yields nine nominal sequences, each with 700 frames, and 1,161 prediction samples. With a fixed random seed, overlapping samples are randomly split into 928 training samples and 233 validation samples. Table II reports the training configuration. The training objective is the per-sample SmoothL1 prediction error:

$$e_t = \frac{1}{Kd}\sum_{k=1}^{K}\sum_{j=1}^{d}\mathrm{SmoothL1}(Y_{t+k,j}, F_\theta(X_t)_{k,j}).$$

After training, the median m_e and interquartile range q_e of nominal prediction errors are estimated from nominal samples. The model parameters θ, the normalizer, and the error statistics are then frozen for all subsequent DT rehearsals. For candidate action π, the raw anomaly uses $\epsilon = 10^{-6}$:

$$R_t^\pi = \max\left(0, \frac{e_t^\pi - m_e}{\max(q_e, \epsilon)}\right).$$

Fig. 1 shows the 200-epoch diagnostic training curve. The training and validation losses decrease rapidly in the early epochs and then enter a slowly improving regime.

D. Capacity and window-length diagnostics

Capacity: prediction accuracy versus model size. The one-factor comparisons in Fig. 2 each use 40 training epochs. Increasing the hidden width from 16 to 32 and 64 reduces

TABLE II
TRAINING CONFIGURATION OF THE NORMAL-DYNAMICS EVALUATOR.

Item	Configuration
Nominal segment	$[0, 14.0)$ s
Target frequency	50 Hz
Past window T	50
Future horizon K	10
Stride	5
Hidden dimension	32
Checkpoint epochs	80
Diagnostic epochs	200
Batch size	64
Optimizer	Adam
Learning rate	0.001
Weight decay	0.0001

validation loss from 0.258 to 0.248 and 0.231, alongside a decrease in the 95th-percentile error. Wider networks thus fit nominal local variations more closely. However, increasing width from 32 to 64 raises the parameter count from 12,712 to 33,416, a factor of 2.63. Width 32 retains a compact model while improving on width 16, balancing accuracy and size rather than minimizing prediction loss alone.

History: longer input is not necessarily better. For history lengths of 25, 50, and 75, validation losses are 0.281, 0.248, and 0.310. Short histories provide less information about local changes; longer histories need not preserve more predictive information after pooling. This nonmonotonic result supports 50 samples within the tested range and motivates matching the history to local dynamics rather than simply extending its coverage.

Horizon: retaining a local evaluation scale. Prediction lengths of 5, 10, and 20 yield validation losses of 0.257, 0.248, and 0.249. The median and 95th-percentile errors remain close for lengths 10 and 20, while parameters increase from 12,712 to 19,312. A 10-sample target therefore provides local-deviation evidence without the larger output head, with little error benefit from further extension in this comparison.

Remark 1 (Prediction error and score calibration). Network configuration affects both the center and tail of the error distribution. Changing capacity or windows therefore requires re-estimating nominal error statistics and freezing them before candidate comparison. Lower prediction loss alone does not imply better recovery decisions.

Algorithm 1 summarizes offline learning. Nominal data determine the feature normalization, predictor parameters, and error statistics. All three remain frozen during candidate evaluation.

II. IMPLEMENTATION OF CANDIDATE-ACTION COSTS

Action selection minimizes $J(\pi)$. For each candidate action $\pi \in \{\texttt{Continue}, \texttt{Return}, \texttt{Land}\}$, the evaluated replay loads its archived DT trajectory. The normal-dynamics evaluator then converts this trajectory into the raw anomaly sequence $R^\pi = \{R_1^\pi, \ldots, R_N^\pi\}$.

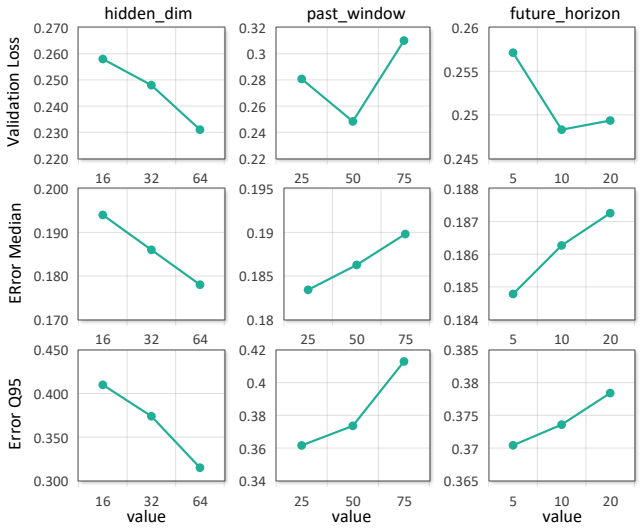


Fig. 2. **Capacity and temporal-window diagnostics.** Validation loss and error statistics under changes in hidden width, history length, and prediction horizon.

Algorithm 2 Candidate-Consequence Cost Ranking

Require: Valid candidate trajectories $\hat{\tau}_\pi$, ordered set $\Pi = (\texttt{Continue}, \texttt{Return}, \texttt{Land})$, frozen evaluator, centers and biases
Ensure: Selected action π^*
1: **for** each candidate action $\pi \in \Pi$ **do**
2: Construct feature windows from $\hat{\tau}_\pi$ and compute raw anomalies R^π using F_θ.
3: Compute $J_{\text{nom}}(\pi)$ from R^π.
4: **end for**
5: Set $S = J_{\text{nom}}(\texttt{Continue})$.
6: **for** each candidate action $\pi \in \Pi$ **do**
7: Compute $P_{\text{sev}}(\pi) = \lambda_s |S - c_\pi|$.
8: Set $J(\pi) = J_{\text{nom}}(\pi) + P_{\text{sev}}(\pi) + b_\pi$.
9: **end for**
10: Select $\pi^* = \arg\min_{\pi \in \Pi} J(\pi)$; break exact ties in the order Continue, Return, Land.

A. Nominal-dynamics cost

The nominal-dynamics cost combines median anomaly, upper-tail deviation, peak deviation, and an endpoint difference:

$$J_{\text{nom}}(\pi) = 0.40\ \text{med}(R^\pi) + 0.30\, Q_{0.9}(R^\pi) + 0.10\ \max(R^\pi) - 0.20\,(R_1^\pi - R_N^\pi).$$

The first three terms capture typical deviation, high-risk tail behavior, and short-term peak risk. The last term rewards a reduction between the first and last valid anomaly values; it is not a fitted trajectory trend. A decreasing endpoint reduces cost.

B. Reference policy labels

Reference labels draw on expert knowledge of UAV emergency handling and on task-retention, return, and landing behavior in fault-flight cases. Under the tested conditions, the accelerometer-noise standard-deviation settings AccStd3, AccStd9, and AccStd15 correspond to Continue, Return, and Land, respectively. This mapping expresses a progression from retaining the mission to returning and terminating flight as degradation increases. Fault-setting identifiers determine labels across trajectories and statistical augmentations, without

TABLE III
PROCESSING OVERHEAD AND POST-COMMAND STATE INTERVALS, EXCLUDING THE REHEARSAL WAIT

Component / case	Measurement	Time (ms)
Neural forward, one action	Mean / p95	0.575 / 0.724
Full evaluation, one action	Mean / p95	0.631 / 0.791
Cost aggregation / selection	Mean / p95	0.673 / 0.741
Three-action compute total	Mean / p95	**2.567 / 2.991**
A9-F8 Return	Marker to home target	363.181
A15-R Return	Marker to home target	96.179
A15-R Land	Marker to mode entry	4.398

TABLE IV
PHYSICAL EXECUTION METRICS OF CANDIDATE ACTIONS IN REAL-FLIGHT CASES.

Case	Action	Duration (s)	Final altitude	Max speed	RMS speed	Peak acc. (m/s^2)
A3-C	Continue	32.19	0.332	0.761	0.507	5.28
A3-C	Return	13.21	-0.041	1.278	0.472	26.34
A3-C	Land	10.09	0.209	0.770	0.387	3.88
A9-F8	Continue	32.09	-0.001	1.405	0.763	20.80
A9-F8	Return	13.40	-0.219	1.690	0.589	21.48
A9-F8	Land	10.29	0.041	0.907	0.406	9.10
A15-R	Continue	24.51	-1.157	1.533	0.640	54.68
A15-R	Return	13.50	0.077	2.252	0.741	46.69
A15-R	Land	10.29	-0.054	1.250	0.626	44.50

deriving labels from S, c_π, J, or selected actions. Ground truth (GT) here denotes the policy-reference label. Labels and centers follow the same intervention ordering; agreement therefore measures policy consistency rather than independent physical safety.

C. Severity reference and final cost

To prevent dynamically smooth but unnecessarily conservative actions from being preferred, the rehearsed consequence of `Continue` is used as the severity reference $S = J_{\mathrm{nom}}(\texttt{Continue})$. The severity-consistency penalty is then

$$P_{\mathrm{sev}}(\pi) = \lambda_s |S - c_\pi|.$$

Reference levels $c^{\mathrm{ref}} = (140, 300, 600)$ are set from the scale of development-scenario Continue costs and the mild, moderate, and severe intervention levels. They are combined with the median Continue cost $\widetilde{S}_\pi$ of each group:

$$c_\pi = \alpha c_\pi^{\mathrm{ref}} + (1-\alpha)\widetilde{S}_\pi.$$

Each group uses three original Continue records, excluding statistical augmentations. With $\alpha = 0.8$ and $\lambda_s = 1$, the group medians and resulting centers are

$$\widetilde{S} = (133.556, 396.601, 652.898),$$
$$c = (138.711, 319.320, 610.580).$$

The reference levels combine the development cost scale with the intervention ordering and are not automatically fitted values. The fixed 4:1 weighting stabilizes the reference scale under limited observations. Values are displayed to three decimal places; scoring uses unrounded values.

TABLE V
PAIRING BETWEEN REAL-FLIGHT CASES AND MAIN-TEXT SCENARIOS. GT DENOTES THE EXPERT POLICY REFERENCE.

Case	Main scenario	GT	DT action
A3-C	VeryLight	`Continue`	`Continue`
A9-F8	TrajectoryShift	`Return`	`Return`
A15-R	Severe-Rect	`Land`	`Land`

$$J(\pi) = J_{\mathrm{nom}}(\pi) + P_{\mathrm{sev}}(\pi) + b_\pi.$$

The fixed biases b_π are 0.00, 0.15, and 0.50 for Continue, Return, and Land. They are not measured mission losses or online task-value inputs. All revised comparisons retain these biases; the selector adds no hard-constraint or cost-history penalty.

Remark 2 (Interpretation of the centers). Continue preserves the current task policy and supplies the common pre-intervention reference. Shrinkage combines intervention reference levels with observed statistics to limit scale fluctuations caused by sparse observations. The calibration defines a decision preference on the anomaly-cost scale, not a physical safety boundary.

Algorithm 2 specifies the scoring stage of the main-text procedure. Replay loads saved candidate trajectories and uses all valid feature windows, computing the three nominal costs before forming S and the final costs. The same candidate order and exact-tie rule apply throughout. Scoring requires valid anomaly sequences for all three candidates.

III. COMMUNICATION, PROCESSING, AND ACTION-EXECUTION LATENCY

This section considers state transmission, scoring and selection, command transmission, and flight-controller response. Table III separates computation benchmarks from onboard event intervals. Compute statistics are recalculated from saved per-run timings. Execution-side intervals use only cases with both a command marker and a subsequent state record.

IV. REAL-FLIGHT CASES AND QUANTITATIVE DETAILS

Table V maps A3-C, A9-F8, and A15-R to VeryLight, TrajectoryShift, and Severe-Rect. Table IV reports duration, final altitude, maximum speed, RMS speed, and peak acceleration for the physical executions.

i) ***Mild degradation: task retention and action transients.*** Continue has the lowest DT cost in A3-C. In the physical records, Return has slightly lower RMS speed than Continue but approximately five times its peak acceleration (26.34 versus 5.28 m/s^2). Lower average motion intensity can therefore coexist with stronger local transients. Ranking an action solely by its conservative intent or an average metric can overlook the dynamics introduced by the intervention itself.

i) ***Moderate degradation: return and immediate landing entail different costs.*** Return has the lowest DT cost in A9-F8. Its RMS speed is lower than Continue's, whereas maximum speed and peak acceleration do not decrease

together; Land has lower maximum speed and peak acceleration. Return should thus not be interpreted as minimizing every physical metric. Its selection combines the intervention level with candidate consequences: the severity reference shapes the intervention preference, while direct costs characterize each action's dynamics, avoiding selection based on a single minimum amplitude.

i) ***Severe degradation: flight termination retains transient loads.*** Land has the lowest DT cost in A15-R. Its physical maximum speed is lower than Return's (1.250 versus 2.252), and its peak acceleration is lower than Continue's (44.50 versus 54.68 m/s^2), but substantial transients remain. Selecting flight termination does not imply immediate removal of dynamic anomalies; peaks and sustained deviations still warrant monitoring during execution.

Across these cases, intervention level and execution smoothness are distinct dimensions. Table IV exposes action-dependent dynamic differences, while Table V connects the cases to DT rankings and policy references. Together they motivate inspecting consequences before execution, without equating individual physical metrics with the final decision cost or safety certification.